

solid border: validated component

NARROWBAND LANE

ALMA science archive:
all public execution
blocks on the 168
census targets within
40 pc (positions
propagated per epoch,
Gaia DR3)

this release: 107 of
208 planned target/bands
(88 stars, 82 systems);
102 EBs; 462 windows
extracted → 431
searched (21 duplicate,
6 defect, 4 withheld)

Doppler-drift narrowband
search at the star
(drift-corrected stack
over a trial drift grid)

same statistic at 512
control positions:
empirical false-alarm
calibration

flag vetting: velocity
mask + known-line
check; recurrence,
closure phase,
cross-target occupancy

**431 windows → 20 $\geq 5\sigma$
→ 4 stage-1 flags
→ 0 candidates;
nominal EIRP_{5 σ} trigger
thresholds per target
(not completeness limits)**

CONTINUUM LANE

line-excluded continuum
at the star position
($\geq 5\sigma$ detection
or 5σ upper limit)

expected photospheric
flux from Gaia
 T_{eff} , radius
(Planck)

population screen:
excess vs same- T_{eff}
bin (median/MAD
outlier test)

**continuum dispositions:
57 measurements (17
detections, 40 limits);
one screen flag,
astrophysically
unsurprising**

validation: injection-recovery on real visibilities (1200 trials; 42% at threshold, 83% at 2x threshold power)
chirp + periodicity re-analysis (122 spectra, demonstration-grade) | cross-target frequency-occupancy check (366 star pairs)